Diagnosing and Repairing Citation Failures in Generative Engine Optimization

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Abstract

Generative Engine Optimization (GEO) aims to improve content visibility in AI-generated responses. However, existing methods measure contribution—how much a document influences a response—rather than citation, the mechanism that actually drives traffic back to creators. Also, these methods apply generic rewriting rules uniformly, failing to diagnose why individual document are not cited. This paper introduces a diagnostic approach to GEO that asks why a document fails to be cited and intervenes accordingly. We develop a unified framework comprising: (1) the first taxonomy of citation failure modes spanning different stages of a citation pipeline; (2) `AgentGEO`, an agentic system that diagnoses failures using this taxonomy, selects targeted repairs from a corresponding tool library, and iterates until citation is achieved; and (3) a document-centric benchmark evaluating whether optimizations generalize across held-out queries. `AgentGEO` achieves over 40% relative improvement in citation rates while modifying only 5% of content, compared to 25% for baselines. Our analysis reveals that generic optimization can harm long-tail content and that some documents face barriers no optimization can overcome—findings with implications for equitable visibility in AI-mediated information access. Code is available at `https://anonymous.4open.science/r/AgentGEO-2D80`.

1. Introduction

The rise of generative engines has sparked significant concern among content creators about traffic loss (The Economist, 2025). Unlike traditional search engines that drive users to click through to websites, generative engines synthesize answers directly, potentially reducing the organic traffic that sustains the creator economy. In response, major platforms have introduced *citation* mechanisms—Google AI Overview displays source links alongside summaries, Perplexity provides inline citations, and ChatGPT's search mode includes expandable references. These citations are now the primary pathway from synthesized answers back to original webpages.

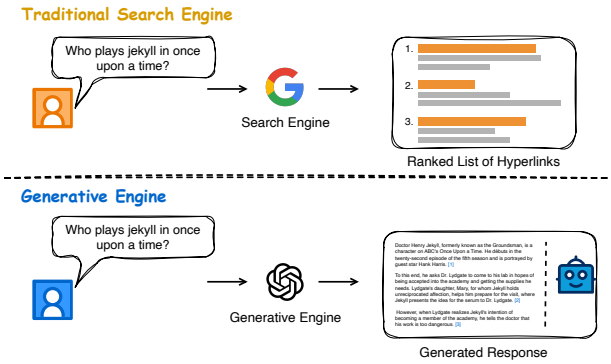

Figure 1. Comparison between traditional search engines and LLM-powered generative engines. Traditional search engines return a ranked list of hyperlinks based on relevance to the user query, while generative engines synthesize answers using retrieved content and provide citations to source webpages.

This pathway, however, is narrow. A recent analysis of 68,879 Google searches reports that only 1% of users who encounter AI summaries click cited sources, compared to 15% who click traditional results when no AI summary appears (Chapekis & Lieb, 2025). Yet the traffic that does arrive converts at substantially higher rates—one analysis found generative search visitors convert 23 times better than traditional organic visitors (Stox, 2025). For content creators, citation is thus the gateway to whatever traffic remains, and a webpage that is not cited receives no referral opportunity regardless of its relevance or quality.

From Contribution to Citation. These dynamics have motivated growing interest in Generative Engine Optimization (GEO): modifying web content to improve visibility in AI-generated responses. Existing work evaluates visibility with *contribution* metrics such as Word Count, which measure the fraction of a response attributed to a source (Aggarwal et al., 2024). While useful, these continuous metrics conflate two factors: whether a webpage is cited at all and, conditional on citation, how much of the response is attributed to

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it. For content creators, the critical barrier is non-citation: no citation means no referral opportunity. We therefore focus on a complementary, citation-first objective that isolates the binary event of being cited. In our experiments, 43% of topically relevant webpages receive no citation under baseline conditions. For these webpages, the question is not "how much am I cited?" but "why am I not cited at all?" Understanding and repairing these failures is an underexplored problem with direct implications for the sustainability of web content creation.

Limitations of Existing Methods. Current GEO methods apply generic preference rules uniformly: add statistics, adopt authoritative tone, improve fluency. This approach fails because citation failures are heterogeneous, spanning multiple pipeline stages. A webpage may fail at *fetching* (malformed HTML, JavaScript dependencies), at *parsing* (content truncated or buried below boilerplate), or at *generation* (missing entities, inferior information density versus competitors). Each failure mode requires a different intervention, yet generic rules cannot diagnose which stage failed, nor can they address upstream failures that occur before text is even processed.

Contribution 1: A Diagnostic Approach to GEO. Motivated by these observations, we reframe GEO from "what generic improvements might help?" to "why did this webpage fail to be cited for the relevant queries?" We instantiate this perspective in AgentGEO, an agentic framework for iterative diagnosis and repair. Given a webpage that fails to be cited, AgentGEO compares it against cited competitors to identify the most likely disadvantage, selects an appropriate tool from a library of specialized interventions spanning the entire citation pipeline, and verifies whether citation is achieved. If not, the system re-diagnoses and iterates, with a memory mechanism to track prior attempts.

Contribution 2: Taxonomy of Citation Failures. We introduce the first systematic taxonomy of citation failure modes spanning the generative engine pipeline: parsing-stage failures (malformed HTML, excessive noise), fetching/context failures (truncation, poor content ordering), and generation-stage failures (entity gaps, intent mismatch, competitor disadvantage). This taxonomy guides our tool library design and provides a foundation for future citation-focused optimization (Figure 2).

Contribution 3: Document-Centric Benchmark. Existing benchmarks pair each document with a single query (Aggarwal et al., 2024; Wu et al., 2025). In practice, content providers cannot anticipate exact queries; they need optimizations that generalize. We introduce MIMIQ (Multi-Intent Multi-Query), a document-centric benchmark associating each webpages with multiple queries spanning diverse intents, personas, and phrasings. This benchmark enables methods to optimize using a training query set and be evaluated on held-out queries, testing whether optimization produces genuinely more citable content rather than overfitting to specific formulations.

Empirical Takeaways. Our experiments yield fresh insights into why webpages fail to achieve citation and what optimization can—and cannot—fix.

(1) Diagnosis beats generic rules while preserving content integrity (Section 6.2). AgentGEO achieves over 40% relative improvement in citation rates across multiple generative engines. Crucially, it does so with minimal modification: baseline methods alter 25% of original content on average, while our targeted repairs touch only 5%. This suggests that citation failure is rarely a global content quality problem: most webpages need targeted fixes, not extensive rewriting.

(2) Generic rules can harm long-tail content (Section 6.3). We identify topic categories where generic optimization not only fails to improve citation but actively degrades it, whereas diagnostic optimization yields consistent gains on the same webpages. This divergence reflects a fundamental limitation of existing methods: generic rules are derived from aggregate patterns, yet specialized topics and underrepresented domains deviate systematically from these patterns. Diagnostic optimization, by conditioning on each webpages's specific failure mode, avoids this bias and thus generalizes more equitably across content types.

(3) Not all citation failures are recoverable (Section 6.4). For certain webpages, even diagnostic optimization fails to improve citation. These webpages face disadvantages—dominant competitors—that no content-side modification can overcome. This finding has implications beyond optimization: if some content is systematically disadvantaged regardless of effort, citation mechanisms may amplify certain voices over others, and creator-side optimization alone cannot ensure equitable visibility.

2. Related Work

2.1. Generative engines

GEs integrate LLMs to retrieve, synthesize, and generate responses directly based on web content utilizing retrieval-augmented generation (RAG) techniques (Lewis et al., 2020; Karpukhin et al., 2020). The workflow of a GE can be roughly divided into three stages: (1) retrieval, (2) content extraction (fetching), and (3) response generation. The retrieval stages identify and retrieve relevant web pages based on user queries. This process may employ traditional search engines (Google, 2025; Willison, 2025) or specialized neural retrieval models (Karpukhin et al., 2020). The content extraction stage focuses on parsing useful text from the retrieved pages, transforming unstructured web content into

structured information suitable for LLM processing. Finally, the response generation stage leverages LLMs to produce coherent, contextually relevant answers derived from the extracted evidence. Beyond these foundational workflows, recent research has advanced more sophisticated GEs by incorporating the reasoning capabilities of LLMs to iteratively plan, reason, and gather evidence for complex queries (Jin et al., 2025; Mo et al., 2025). These developments have broadened the research focus to encompass not only the integration of retrieval and generation but also the improvement of factual consistency and reliability in responses (Salemi & Zamani, 2024; Niu et al., 2024).

2.2. Search Engine Optimization

Search engine optimization (SEO) has long been a pivotal area in information retrieval (Davis, 2006; Yalçın & Köse, 2010). Traditional SEO strategies focus on boosting webpage rankings in a list of search results through techniques such as keyword optimization, backlink management, and site structuring (Killoran, 2013). With the search paradigm shift toward GEs, GEO emerges to enhance the visibility of content within generated responses. Early work has demonstrated the efficacy of leveraging LLMs to rewrite web content based on manually crafted heuristics to improve retrieval likelihood (Aggarwal et al., 2024; Chen et al., 2025a). To better align with GE preferences, recent work leverages LLMs to distill optimization rules for text rewriting (Wu et al., 2025). Beyond content optimization, researchers have also investigated adversarial techniques, such as prompt injection attacks (Nestaas et al., 2024; Pfrommer et al., 2024; Tang et al., 2025), to manipulate GE behaviors. Despite these advancements, existing approaches predominantly focus on content-level textual modifications. They often neglect critical structural and systemic factors, such as HTML parsing mechanisms, context window constraints, and webpage structure, which significantly influence how content is ingested by the engine. In contrast, `AgentGEO` presents a holistic GEO framework that addresses these limitations by optimizing across multiple aspects of the webpage representation.

2.3. GEO Benchmarks

GEO-Bench (Aggarwal et al., 2024) is one of the first GEO benchmarks, which is designed for query-centric evaluation. The training set contains queris from diverse sources and test set includes both queries and associated retrieved documents. It evaluate GEO methods by measuring the improvement of one documents's visibility against others for the query after optimization. E-commerce (Wu et al., 2025) and Researchy-GEO (Wu et al., 2025) have similar settings but focus on specific domains. CC-GSEO-Bench (Chen et al., 2025b) is another GEO benchmark that focuses on document-centric evaluation, where each document is associated with multiple queries. However, the number of queries per document is limited and inconsistent, ranging from 2 to 10, which makes it difficult to comprehensively assess a document's overall visibility in GEs. All these benchmarks only consider content and cannot evaluate HTML structure data. In contrast, our **MIMIQ** evaluates both content and HTML structure, enabling a more complete assessment of GEO strategies.

3. Understanding Citation Failures

We adopt citation—the binary event of being included as a source—as our optimization target. To understand why documents fail to be cited, we first examine how generative engines implement citation, then present a taxonomy of failure modes.

3.1. Citation Mechanisms in Generative Engines

How a generative engine attributes its response to sources determines where citation failures can occur. We identify the follwing three paradigms.

Inline Attribution. Reference markers (e.g., "[1]") are embedded directly within the generated sequence, providing fine-grained provenance at the sentence or clause level. The model decides *during generation* which sources support each claim. Systems like Perplexity and ChatGPT's search mode employ this approach (Perplexity, 2026; OpenAI, 2024).

Pre-hoc Attribution. The system first selects passages from retrieved documents, then generates a response grounded exclusively in these selections. Because generation operates over pre-selected evidence, the source context serves as an inherent anchor—the model cannot easily hallucinate beyond what was extracted (Slobodkin et al., 2024). However, relevant content not selected upstream will not be cited.

Post-hoc Attribution. Generation and verification are decoupled: the model first produces a response, then a separate module retrospectively identifies supporting sources (Bohnet et al., 2022; Gao et al., 2023). While flexible, this paradigm risks hallucination propagation—errors introduced during generation may persist if the verifier incorrectly validates unsupported claims.

Given their prevalence in commercial systems and tighter grounding guarantees, we focus on inline and pre-hoc attribution. These paradigms expose different failure surfaces: inline attribution can fail when the model overlooks a relevant source during generation, while pre-hoc attribution can fail earlier, during passage selection. Our taxonomy accounts for failures across both paradigms.

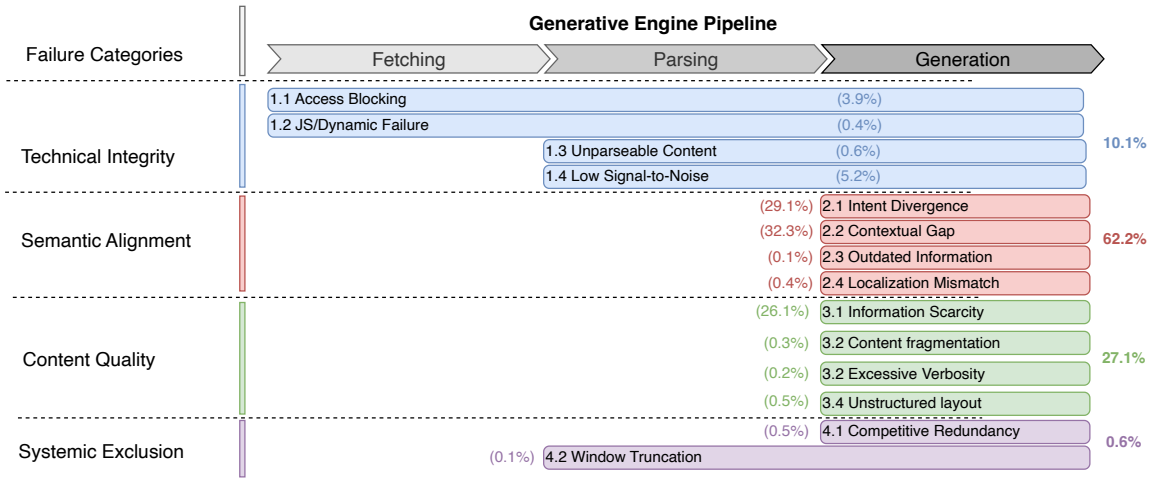

Figure 2. A taxonomy of citation failure modes in generative engines, spanning html fetching, prasing and answer generation stages.

3.2. A Taxonomy of Citation Failures

Why does a topically relevant webpage fail to be cited? To answer this systematically, we construct a diagnostic dataset from GEO-Bench (Aggarwal et al., 2024), which spans 10 diverse domains (health, travel, technology, etc.) and includes queries with varying intents. For each query where a retrieved webpage was not cited, we pair it with a cited competitor, yielding 949 contrastive pairs. This pairing strategy isolates the marginal factor: both webpage were retrieved for the same query, so the difference lies in why the engine preferred one over the other. By analyzing these pairs, we identify four primary failure dimensions mapped across the generative engine pipeline (Figure 2). Detailed explanation of each category is provided in Appendix C.

Technical Integrity (10.1%). The webpage cannot be properly ingested. Failures occur at the *fetching* stage (access blocking, JavaScript rendering failures, connection errors) or the *parsing* stage (unparseable content, low signal-to-noise ratio from boilerplate, formatting fragmentation). A webpage with critical information buried in a JavaScript widget or overwhelmed by navigation chrome may never surface its content to the generator.

Semantic Alignment (62.2%). The webpage's content does not match what the query requires. This includes *intent divergence* (informational content for a transactional query), *contextual gaps* (missing specific entities or terminology), *outdated information*, and *localization mismatch* (e.g., UK regulations for a US query). These are generation-stage failures: content reaches the model but is judged insufficiently relevant.

Content Quality (27.1%). The webpage addresses the right topic but presents information poorly. Manifestations include *information scarcity* (too shallow to cite), *content fragmentation* (disconnected snippets resisting synthesis), *excessive verbosity* (key facts diluted by filler), and *unstructured layout* (dense prose where tables would aid extraction). These are generation-stage failures where better-presented competitors win.

Systemic Exclusion (0.6%). The webpage faces structural disadvantages beyond its content. *Competitive redundancy* occurs when a higher-authority source (e.g., Wikipedia) covers identical facts. *Window truncation* occurs when relevant content is buried too deep to fit the context window.

4. AgentGEO: A Diagnostic Approach to GEO

Problem Setup. Let d denote a target webpage and $\mathcal{Q}$ a distribution of user queries relevant to d. For query q, let $\mathcal{V}(q, d, \mathcal{G}) \in \{0, 1\}$ indicate whether webpage d is cited by generative engine $\mathcal{G}$. We seek an optimized webpage d^* that maximizes expected citation across $\mathcal{Q}$ while preserving fidelity to the original:

$$d^* = \arg\max_{d'} \; \mathbb{E}_{q \sim \mathcal{Q}} \left[\mathcal{V}(q, d', \mathcal{G}) \right] \quad \text{s.t.} \quad \mathrm{Sim}(d, d') \geq \tau$$

In practice, we optimize using training queries Q_{tr} and evaluate on held-out queries Q_{te}.

Overview. AgentGEO follows a three-stage pipeline (Figure 3). First, for each training query, a *diagnose-then-repair* cycle identifies why the target webpage fails to be cited and applies targeted fixes, guided by the taxonomy and a memory module that tracks prior attempts. Second, a *batch aggregation* stage merges query-specific suggestions into query-invariant modifications, preventing overfitting to individual queries. Third, a *localized editing* mechanism applies modifications at the chunk level, preserving semantic integrity.

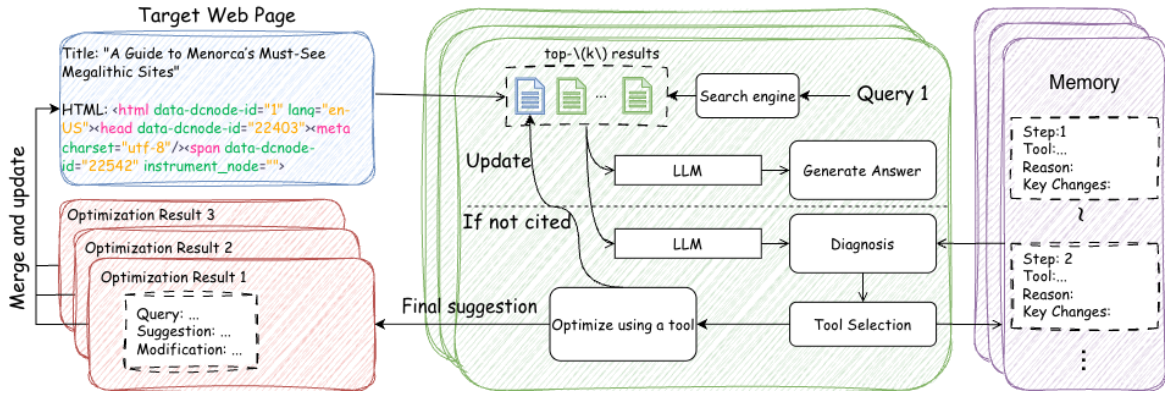

Figure 3. The workflow of `AgentGEO`.

4.1. Diagnose-then-Repair Loop

For each training query q, AgentGEO executes an iterative cycle that transforms the target webpage d until it achieves citation or the iteration limit is reached.

Failure Diagnosis. We simulate the GE pipeline: retrieve candidate webpage, generate a response, and record which sources are cited. If the target webpage is not cited, diagnosis begins. To identify the failure reason, the system compares d against the highest-ranked cited competitor and characterizes the vulnerability, i.e., why the competitor was cited instead. These vulnerabilities are classified according to our taxonomy of failure modes (Section C).

Tool Selection with Memory. Given the diagnosed vulnerability, `AgentGEO` selects a repair tool from a library of specialized interventions spanning HTML repair, content reordering, entity enrichment, and information density improvements (full list in Appendix D). To avoid repeating ineffective repairs, we maintain a query-specific memory that tracks prior attempts. At each iteration t, the agent selects a tool based on the current vulnerability and memory. Memory is initialized empty.

Iterative Refinement. The selected tool is applied to produce a modified webpage $d^{(t)}$. If citation is achieved, the system extracts a suggestion summarizing the successful edits—specifically, which content locations were modified and how. These suggestions feed into batch aggregation described next. If citation is not achieved, memory is updated with the failed attempt, and the system re-diagnoses against competitors. This loop continues until citation succeeds or the iteration limit is reached.

4.2. Aggregating Across Queries

To achieve generalization across training queries, AgentGEO performs webpage updates at the batch level. For each query in a batch, we run the diagnose-then-repair loop (Section 4.1) on an isolated, temporary copy of the current HTML. Each run yields at most one candidate edit, consisting of the diagnosed failure type, a target structural chunk index, and a tool-produced HTML fragment to update that chunk. To keep edits comparable across queries, we fix the chunk partition at the start of the batch and map all candidate edits into this frozen index space, even if the temporary DOM changes during iterative retries. We aggregate candidate edits by target chunk and, for each chunk, select at most one edit to apply to the shared webpage. Our default diagnosis-aware selector ranks candidates by failure severity and a confidence score, resolving conflicts by retaining the highest-ranked proposal. (Other conflict-resolution strategies include heuristic voting within a chunk or an LLM-based arbiter.) The selected per-chunk edits are applied in a single pass to synthesize the batch update, mitigating query-specific overfitting and promoting query- invariant improvements. We do not enforce an explicit similarity threshold at aggregation time; fidelity is primarily maintained through localized chunk-level editing (Section 4.3).

4.3. Localized Editing for faithfulness

To satisfy the semantic preservation constraint, AgentGEO applies modifications at the chunk level rather than rewriting entire webpages. This serves two purposes: it preserves faithfulness by restricting edits to targeted locations, and it avoids the quality degradation LLMs exhibit when processing long webpages (Liu et al., 2024).

The webpage is partitioned into structure-aware chunks using HTML tags, where each chunk represents a coherent unit (e.g., a paragraph, list, or metadata block). Each suggestion from the aggregation stage specifies the content location it targets; we map these locations to the corresponding chunks. Modifications are applied only to matched chunks; the remaining content and overall HTML structure remain unchanged.

Table 1. Comparison of **MIMIQ** and existing GEO benchmarks. Our dataset uniquely combines a document-centric paradigm with a holistic evaluation scope and supports explicit train/test query splits.

Dataset	Doc-centric	HTML	Train/Test Split
GEO-Bench (Aggarwal et al., 2024)	×	×	✓
E-commerce (Wu et al., 2025)	×	×	✓
Researchy-GEO (Wu et al., 2025)	×	×	✓
CC-GSEO-Bench (Chen et al., 2025b)	✓	×	×
MIMIQ (Ours)	✓	✓	✓

5. Document-Centric Benchmarking

Existing GEO benchmarks pair each document with a single query and evaluate optimization on that same query (Aggarwal et al., 2024). This setup risks conflating genuine improvements with query-specific overfitting: an optimization that succeeds for one query may fail for another expressing the same intent.

We introduce **MIMIQ** (**M**ulti-**I**ntent **M**ult**I**-**Q**uery), a document-centric benchmark for evaluating generalization. We call it *document-centric* because it treats the document—not the query-document pair—as the unit of optimization and evaluation. This mirrors the content creator's setting: they control their document but cannot anticipate the exact queries users will issue. For each document, MIMIQ provides multiple queries spanning diverse intents, personas, and phrasings. Methods optimize using a training query set and are evaluated on held-out queries unseen during optimization. This protocol tests whether an optimization produces genuinely more citable content—or merely overfits to specific query formulations. Table 1 compares **MIMIQ** to existing benchmarks.

5.1. Construction Pipeline

We use LLM to generate a dense set of queries for each target webpage. To generate queries that reflect realistic search diversity, we design a *structured* generation framework spanning three dimensions: intent, persona, and phrasing. Detailed prompt templates and filtering heuristics are provided in Appendix A.

Content-Anchored Keyword Extraction. Directly utilizing entire web page content for query generation often introduces noise. For example, extracting keywords from a page about "safe road trips" may yield irrelevant terms like "maps". To maintain relevance, we extract keywords exclusively from the page `title` metadata, which typically encapsulates the core topic concisely. We instruct the LLM to derive three categories of terms: (1) *Core Keywords* for topical relevance, (2) *Synonyms/Latent semantic indexing terms* for semantic expansion, and (3) *Long-tail Phrases* to capture specific user needs.

Persona and Intent Modeling. To mimic heterogeneous user behaviors, we incorporate personality simulation and intent classification. We synthesize diverse user personas (e.g., "Domain Expert" vs. "Casual Browser") and map them to standard search intents (Navigational, Informational, Transactional, Commercial) (Broder, 2002; Rose & Levinson, 2004). For each webpage, we filter out implausible intents (e.g., removing navigational intent for a pure recipe page) to ensure the generated queries are contextually valid.

Quality Control. The raw generated queries undergo an additional deduplication and relevance filtering process. We employ a verifier model to eliminate hallucinations, ensuring that every query in the final set is logically answerable by the target webpage.

5.2. Dataset Statistics and Variants

The standard MIMIQ dataset comprises 204 webpages sampled from the ClueWeb22 index. For each webpage, we curate a stratified set of 60 queries, split into 20 for training and 40 for testing, resulting in a total of 12,240 queries. Beyond the standard setting, we introduce two challenging variants to probe specific optimizer capabilities:

- **MIMIQ-OOD (Out-of-Distribution):** To evaluate generalization, we enforce a distributional shift between training and testing. We cluster generated queries based on user personas; specific personas are reserved exclusively for the training set, while unseen personas constitute the test set. This setting tests whether an optimizer can generalize to unseen user behaviors.
- **MIMIQ-HTML (Structural Robustness):** In practice, citation failures often originate in the upstream fetching and parsing phases. MIMIQ-HTML subset identified as having complex or irregular DOM structures. This variant specifically evaluates the optimizer's ability to parse and modify raw HTML.

Each variant comprises 50 webpages with 60 queries per page, and maintaining the same train/test split as the standard dataset.

6. Experiment

In this section, we conduct extensive experiments to evaluate the effectiveness of `AgentGEO` on proposed datasets. We first introduce the experimental settings, including baselines and evaluation metrics. Then we present the main results and analyze the impact of OOD data. Finally, we perform ablation studies to validate the contribution of each component in `AgentGEO`.

Table 2. Performance comparision of AgentGEO againse baselines across different citation methods. **Bold** and underline indicate the best and second-best results, respectively. "-" indicates the metric is not applicable to the method.

	In-context							Attribute first then generate						
	CR	Contribution			Faithfulness			CR	Contribution			Faithfulness		
Method	**CR**	**Word**	**Pos**	**Wordpos**	**TF-IDF**	**Embed**	**Jaccard**	**CR**	**Word**	**Pos**	**Wordpos**	**TF-IDF**	**Embed**	**Jaccard**
Vanilla	56.58	16.28	16.28	16.39	-	-	-	60.20	23.69	28.72	23.57	-	-	-
Technical Terms	55.84	14.78	14.73	14.60	73.32	85.64	21.37	57.33	22.25	27.87	22.03	74.39	86.98	21.98
Cite Sources	60.52	17.01	16.96	16.89	92.76	93.62	65.62	59.21	24.04	29.02	23.79	92.33	92.95	64.60
Keyword Stuffing	62.44	17.83	17.74	17.82	82.85	88.95	30.04	65.62	28.09	33.39	27.81	82.16	86.92	28.39
Unique Words	57.76	15.33	15.37	15.19	86.00	91.58	32.72	57.42	23.43	28.45	23.23	86.11	91.58	33.30
Authoritative	59.14	17.45	17.75	17.57	80.42	86.38	25.15	63.46	26.86	31.85	26.63	80.85	86.35	26.02
Easy-to-Understand	71.11	19.73	19.84	19.75	82.17	86.68	30.26	65.34	28.44	33.62	27.98	81.99	86.93	30.41
Statistics Addition	60.52	17.41	17.29	17.26	95.52	**94.19**	69.31	59.46	24.15	29.39	23.93	95.51	93.75	69.38
Quotation Addition	59.00	16.28	16.16	16.19	**96.78**	93.92	74.85	59.99	23.66	28.88	23.48	96.57	**94.29**	74.78
Fluency Optimization	60.60	16.52	16.63	16.48	92.40	92.72	48.98	62.22	25.44	30.95	25.25	92.20	92.83	49.21
AutoGEO	68.80	17.87	18.53	17.65	67.49	76.99	17.97	65.97	33.96	**34.33**	**32.88**	74.55	78.52	19.77
AgentGEO (ours)	**79.52**	**20.29**	**21.08**	**20.28**	94.23	93.20	**82.40**	**70.00**	**34.18**	31.57	30.92	**97.44**	92.77	**82.66**

Table 3. Performance comparison across different LLM-based generative engines. **Bold** indicates the best performance.

Model	Metric	Vanilla	AutoGEO	AgentGEO (ours)
GPT	CR	56.58	68.80	**79.52**
	Word	16.28	17.87	**20.29**
	Pos	16.28	18.53	**21.08**
	Wordpos	16.39	17.65	**20.28**
	Precision	48.63	56.79	**58.43**
	Recall	64.62	**68.99**	67.10
	Clarity	61.23	**66.08**	62.88
	Insightfulness	40.20	**45.25**	42.68
	TF-IDF	-	67.49	**94.23**
	Embed	-	76.99	**93.20**
	Jaccord	-	17.97	**82.40**
Claude	CR	42.40	46.80	**54.80**
	Word	21.80	24.86	**26.94**
	Pos	21.54	24.81	**27.57**
	Wordpos	21.44	24.66	**26.10**
	Precision	89.42	88.79	**90.33**
	Recall	**96.16**	95.51	96.02
	Clarity	73.99	**75.74**	74.99
	Insightfulness	57.47	**61.05**	60.76
	TF-IDF	-	74.27	**95.26**
	Embed	-	79.98	**88.76**
	Jaccord	-	22.28	**74.58**

6.1. Experimental Settings

Baselines We consider three types of baselines in our experiments. Vanilla baseline is the original generative engine without any optimizating. GEO-Bench (Aggarwal et al., 2024) propose eight methods to optimize website content using LLM. AutoGEO (Wu et al., 2025) automatically summarize preference rules from a large number of webpage paires and optimize the website content accordingly. We test all the methods on generative engines built with state-of-the-art LLMs, including GPT (gpt-4.1-mini), and Claude (claude-haiku-4-5-20251001).

Metrics. We evaluate model performance along with three dimensions: *visibility*, *utility* and *webpage faithfulness*. Visibility quantifies the extent to which optimized content is surfaced and prioritized by GEs in response. We calculate the **citation rate** (CR), which measures the proportion of optimized webpages successfully cited in test queries. As mentioned in the Introduction, this will be the primary metric we optimize for. We also report three other metrics from GEO-Bench (Aggarwal et al., 2024): **Word, Pos, and Wordpos**, which assess the structural and lexical contribution of the content. Utility assesses the quality of the generated answers. We follow DeepResearchGym (Coelho et al., 2025) to evaluate utility, covering relevance (**Precision, Recall**), and quality (**Clarity, Insight**). Faithfulness evaluates the retention of original information in the optimized webpages. We quantify this using three metrics: **TF-IDF, Embedding similarity, and Jaccard similarity**. TF-IDF computes the cosine similarity of weighted term vectors, whereas Embedding similarity utilizes dense representations from pre-trained language models to assess semantic proximity. Finally, Jaccard similarity measures the strict lexical overlap between the webpages' unique word sets. All results are reported as percentage values (%).

GE Simulation Setup. Since production GE pipelines are proprietary, we construct a controlled simulation using standard retrieval and generation components with explicit citation instructions. This enables systematic diagnosis, reproducible evaluation, and controlled ablations that would be infeasible with black-box commercial systems.

6.2. Overall GEO performance

Performance across different citation methods. We first compare AgentGEO with different baseline citation methods. We evaluate on two citation methods, including in-context generation and attribute-first-then-generate (Slobodkin et al., 2024). In in-context generation, the generative engine is directly prompted to generate answers with citations based

on the optimized webpage. In attribute-first-then-generate, the generative engine first extracts citation-related attributes from the optimized webpage and then generates answers based on these attributes.

Table 2 presents the results. We observe that AgentGEO consistently outperforms all baselines across different citation methods and generative engines, demonstrating its effectiveness and robustness in enhancing citation performance. In in-context generation, AgentGEO achieves a citation rate of 79.52%, outperforming the best baseline, AutoGEO, by 10.72%. In attribute-first-then-generate, AgentGEO attains a citation rate of 70.00%, surpassing AutoGEO by 4.03%. Additionally, AgentGEO also excels in contribution and faithfulness metrics, indicating that it not only improves citation rates but also maintains high-quality and faithful content.

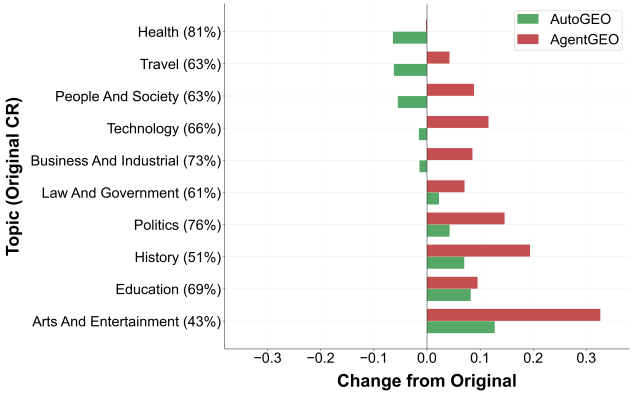

Figure 4. Citation rate improvement over vanilla baseline across different topics. The numbers in left indicate the original citation rate for each topic.

Performance across different LLM-based generative engines. We further evaluate whether AgentGEO's advantages hold across different GEs. For efficiency, we evaluate on a subset of 50 webpages sampled from the standard dataset. Due to computational constraints, we evaluate on a subset of 50 webpages sampled from the standard dataset. Table 3 presents the results with GPT and Claude. AutoGEO and our method both have a better citation rate on GPT than on Claude. One possible reason is that GPT has a stronger ability to understand and leverage optimized content for citation purposes, while Claude may require more explicit modifications to the content to achieve similar citation improvements. However, when evaluating utility metrics such as Precision, Recall, Clarity, and Insightfulness, Claude generally outperforms GPT across all methods. This suggests that Claude may have a stronger capability for generating high-quality and relevant answers, regardless of the optimization method used. Meanwhile, on both models, AgentGEO has comparable performance to Vanilla and AutoGEO on utility metrics, indicating that our optimization does not compromise answer quality while enhancing citation rates.

6.3. Evaluation of different topics.

To analyze the effectiveness of AgentGEO across different topics, we categorize the webpages into various topics such as arts, education, history, etc. We then evaluate the citation rate improvement over vanilla baseline for each topic. We also compare AgentGEO with AutoGEO as a reference. The results are summarized in Figure 4.

The results show that different topics exhibit varying degrees of citation rate improvement. Topics such as arts and entertainment, history, and politics show significant improvements. However, topics like health show negative or minimal improvements, indicating that AgentGEO may be less effective in these areas. We take a closer look at the health topic, which shows a negative improvement with AgentGEO. We find that the original citation rate for health-related webpages is already quite high, around 80%, leaving limited room for further enhancement. For the remaining uncited webpages, some optimization steps inadvertently removed domain-specific information, making them less likely to be cited. Surprisingly, AutoGEO shows worse performance than the vanilla baseline in several topics. In particular, when the original citation rate is already high, AutoGEO tends to decrease it, suggesting that generic rules may harm long-tail content.

6.4. Failure Optimization Case Study

To understand the limitations of AgentGEO, we analyze failure cases from the 50-webpage optimization experiment. While the overall citation rate improves from 57.0% to 83.7% on training queries, 163 queries remain uncited after optimization. A closer examination of these failures reveals that the optimization process does, in fact, succeed in modifying webpage content to address diagnosed issues. Yet the GE still does not cite these webpages, indicating that the failures stem not from ineffective optimization but from the inherent difficulty of aligning webpage content with GE citation criteria.

For example, a university course page on machine learning fails to be cited in response to queries about "best online machine learning courses." The optimization process successfully enhances the page's clarity and relevance by adding detailed course descriptions and student testimonials. Nevertheless, the GE continues to overlook it, likely because it favors authoritative sources such as well-known online education platforms (e.g., Coursera, edX) or prestigious institutions (e.g., Stanford, MIT) offering comparable courses. This suggests that even well-optimized content may struggle to gain visibility when it lacks the perceived authority or credibility that GEs prioritize in their citation mechanisms

— a phenomenon we term *authority competition*. A deeper analysis of such cases could yield valuable insights into the fundamental limitations of content optimization, which we leave for future work.

7. Conclusion

This paper presents a diagnostic approach to Generative Engine Optimization that shifts focus from generic content rewriting to understanding why individual webpage fail to be cited. We introduce a taxonomy of citation failure modes, `AgentGEO` —an agentic system that diagnoses failures and applies targeted repairs—and a webpage-centric benchmark for comprehensive evaluation. Future work could focus on validating `AgentGEO` against commercial engines and improving scalability through distillation and efficient diagnosis strategies.

Impact Statement

This research introduces `AgentGEO`, a framework for Generative Engine Optimization (GEO). While GEO empowers content providers to remain visible in an evolving search landscape, it also introduces several potential risks that warrant ethical consideration:

- Information Integrity and Manipulation: The ability to systematically optimize content for LLM citations could be exploited to prioritize biased, low-quality, or even misleading information. If malicious actors use GEO to ensure their content is cited more frequently than authoritative sources, the reliability of generative engines as a primary information gatekeeper could be severely compromised.

- Degradation of Human-Centric Content: Similar to the historical "SEO-spam" in traditional search, a widespread adoption of GEO may incentivize creators to write for "engines" rather than humans. This could lead to a web ecosystem filled with "LLM-friendly" but redundant or stylistically unnatural content, ultimately degrading the quality of information available to human readers.

To mitigate these risks, we emphasize that GEO research should proceed in tandem with the development of robust, fact-checking generative engines. By introducing **MIMIQ** (MIMIQ), we encourage the community to evaluate GEO not just through citation gains, but through the lens of document faithfulness and utility, ensuring that optimization serves to enhance, rather than distort, the global information ecosystem.

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